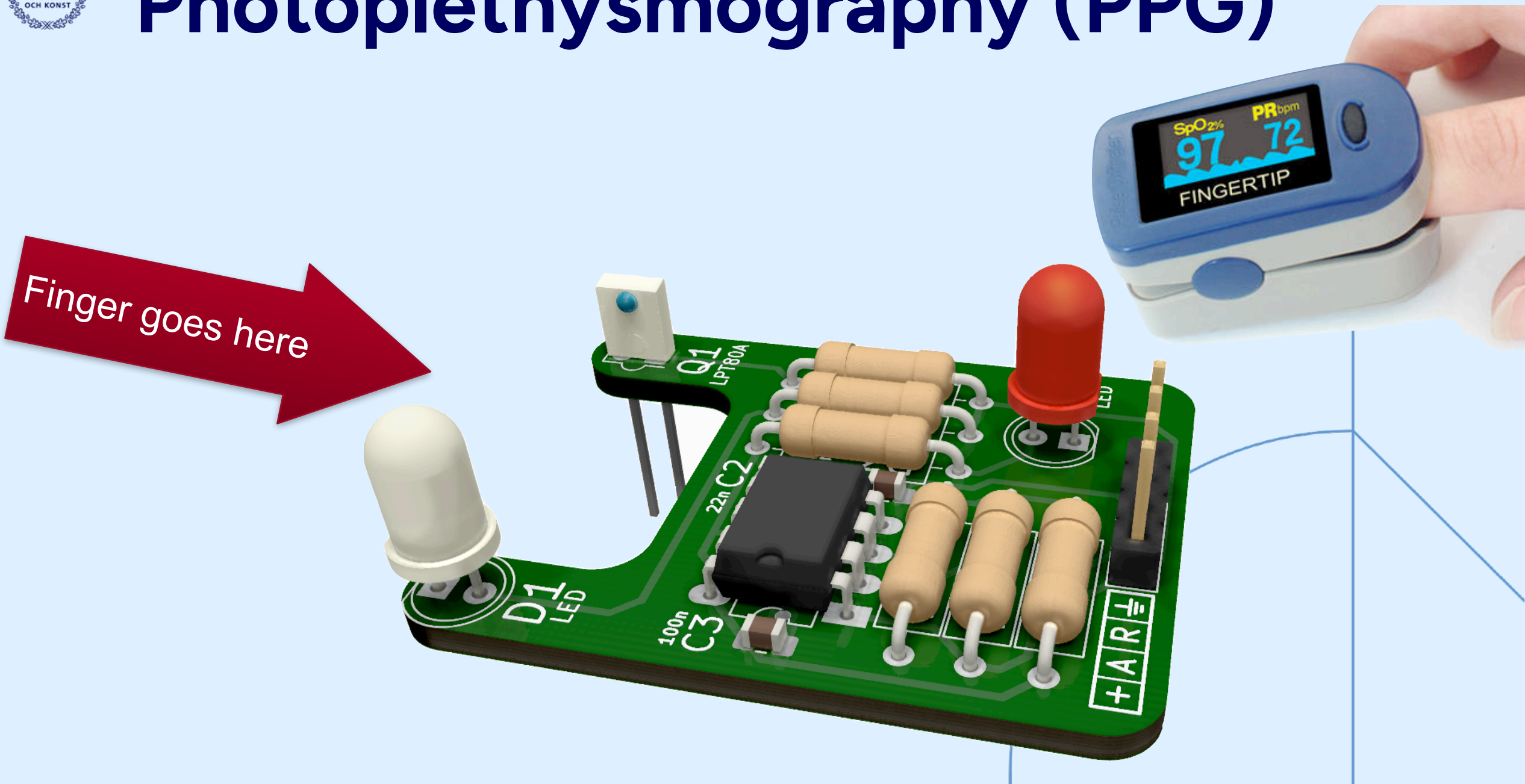


PCC1: Lab 0

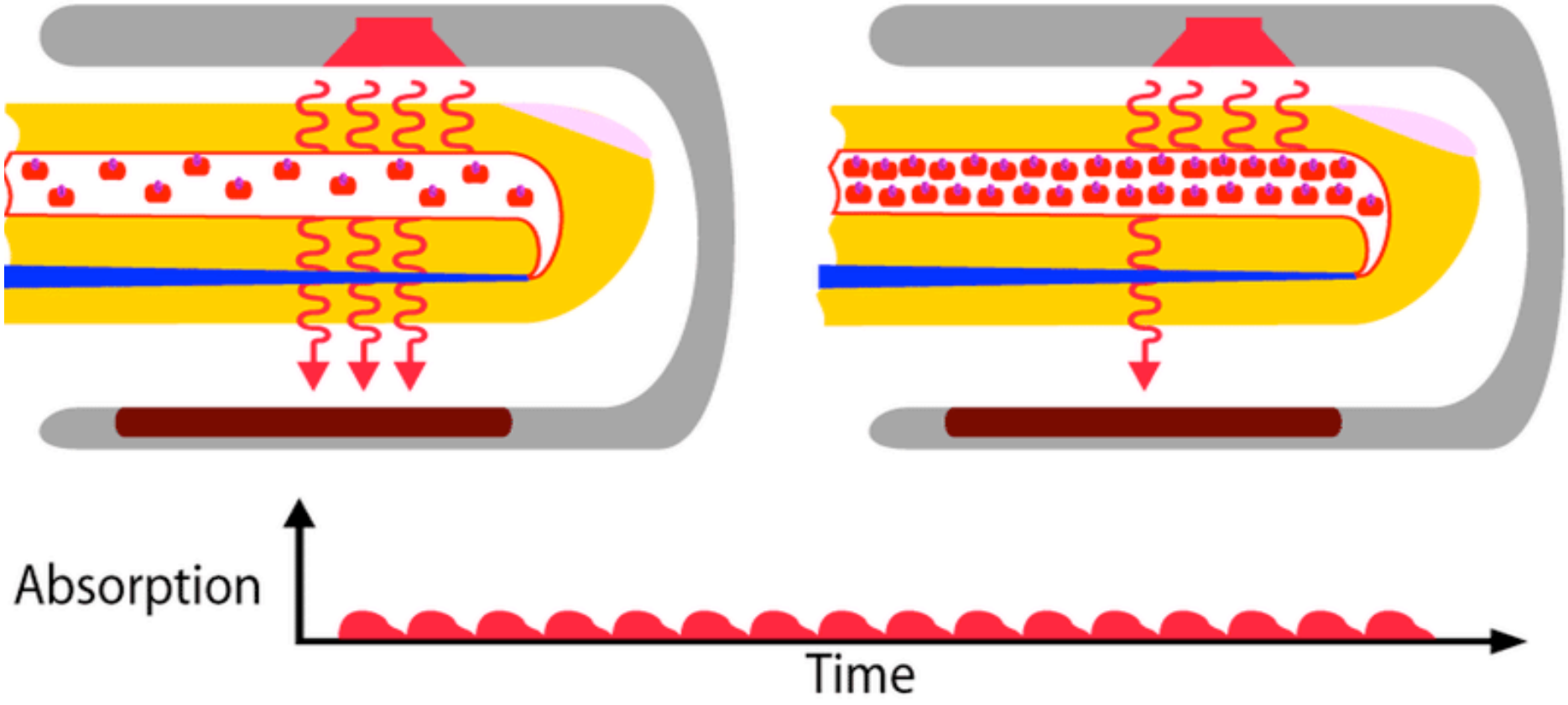
Connect the sensor on breadboard

CM2015 - Martin Jacobsson

Photoplethysmography (PPG)



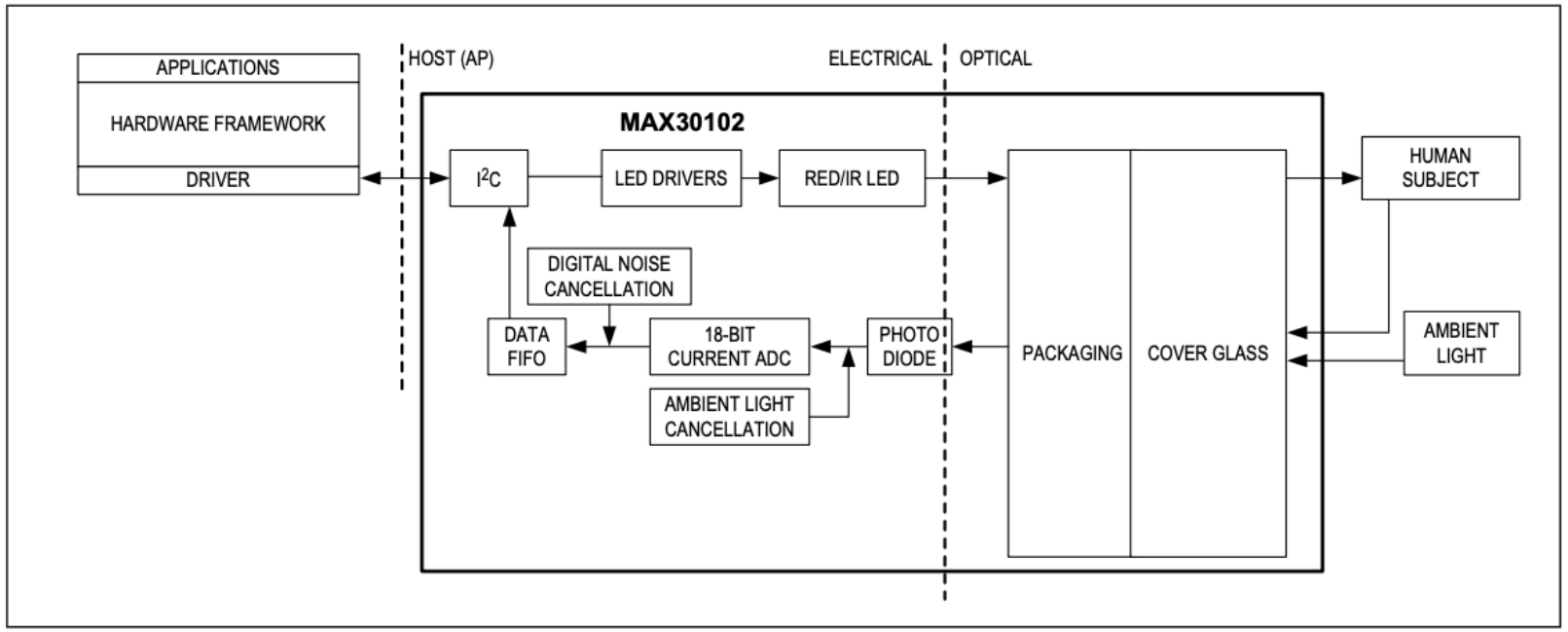
How a PPG sensor works

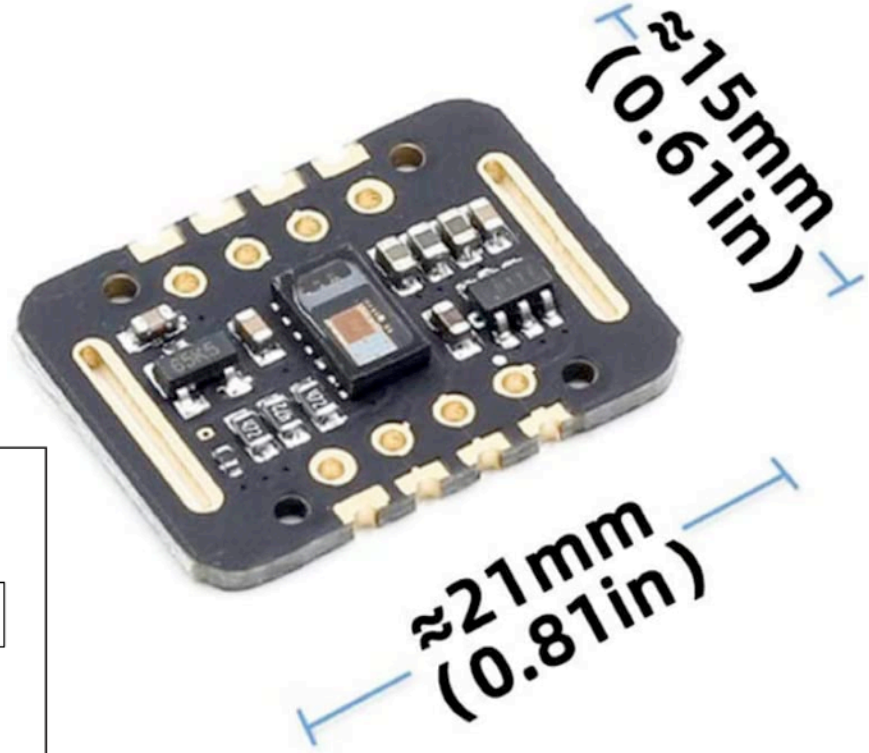


Li, Xiaobai & Komulainen, Jukka & Zhao, Guoying & Yuen, Pong-Chi & Pietikainen, Matti. (2016). Generalized face anti-spoofing by detecting pulse from face videos. 4244-4249. 10.1109/ICPR.2016.7900300.

Maxim MAX3010x

All in one package solutions exists!
Used in many medical SpO_2 devices





maxim integrated™

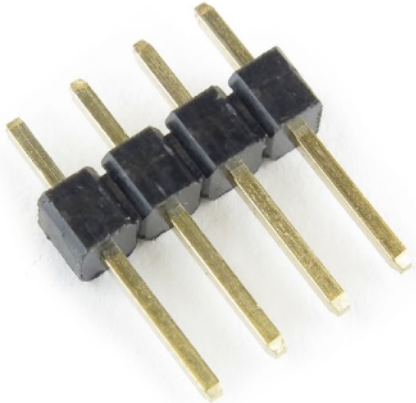
Basic Circuit



- Infrared LED: Vishay TSHA5200
- Infrared Phototransistor: Osram SFH310
- Band Pass Filter with Amplification
- Indicator LED
- Connectors





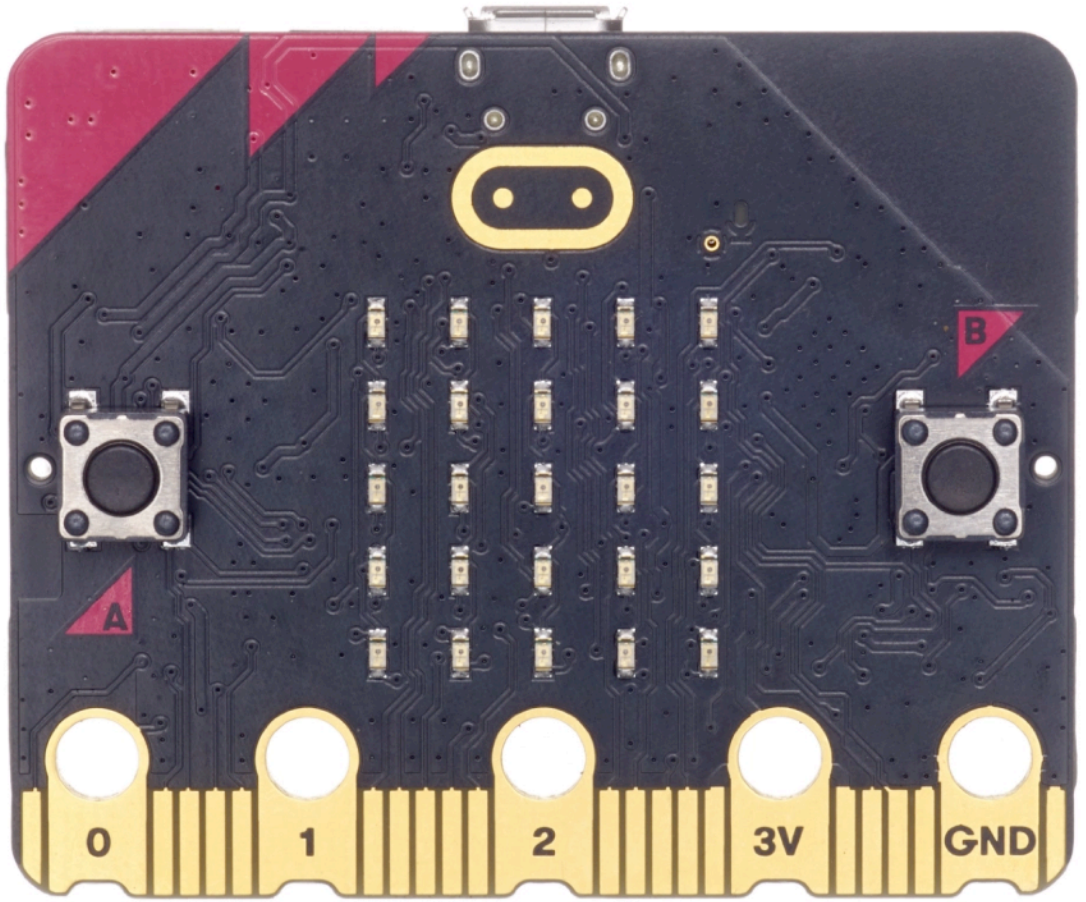




Connect to PPG circuit

- Connect your PPG circuit to:
- 3V
- GND
- PIN0

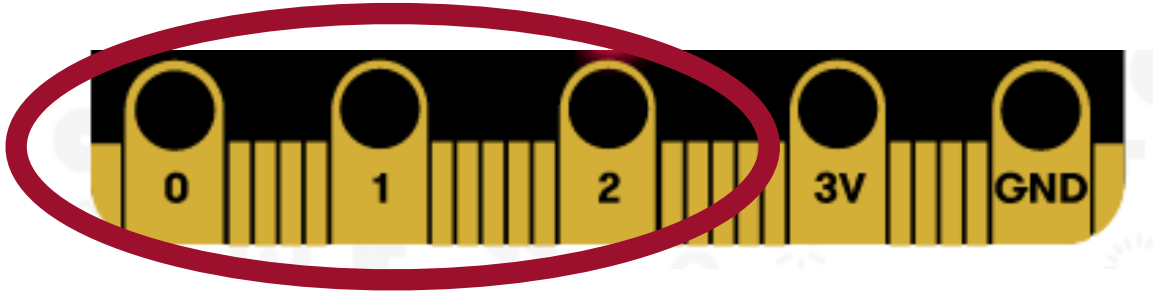
- *PIN0 is an ADC*



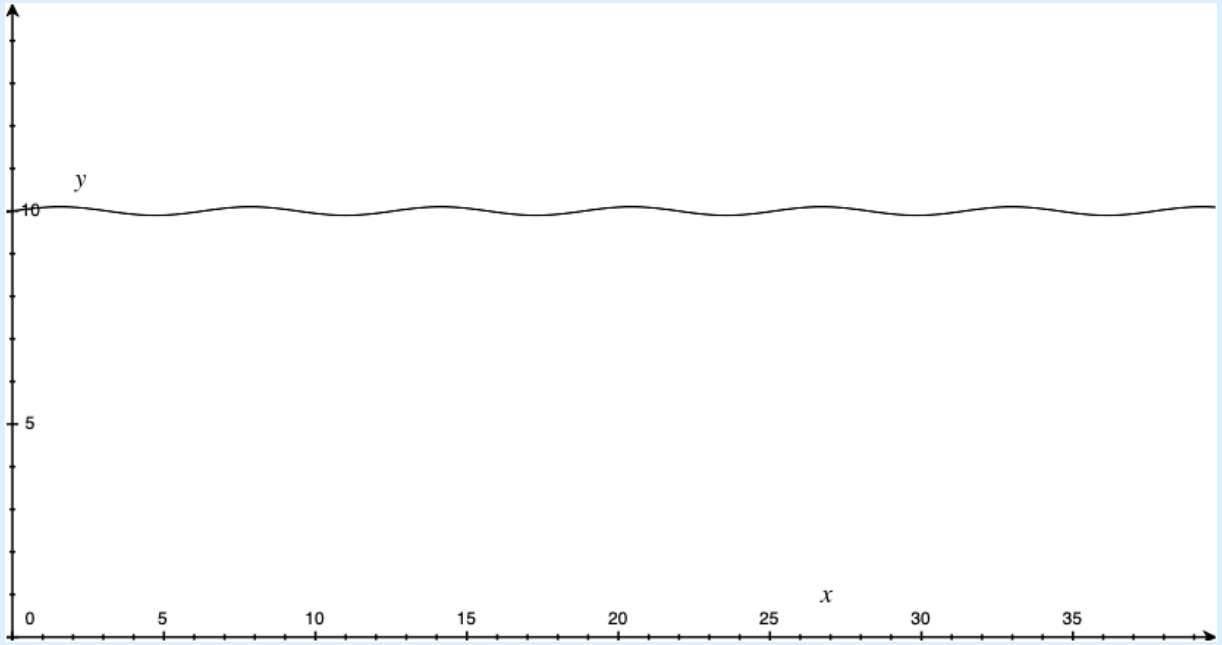
Analog to Digital Converter (ADC)



- Small and fast digital voltage meter
- Measures voltage in an interval (0-3 Volt)
- Outputs an integer value to the software
  - 0 - 1023 (10 bits), 3V ≈ 1023
- Can measure several times per second
  - BBC Micro:bit with software: 400 Hz or faster
- Typical embedded processors have 10 or more ADCs

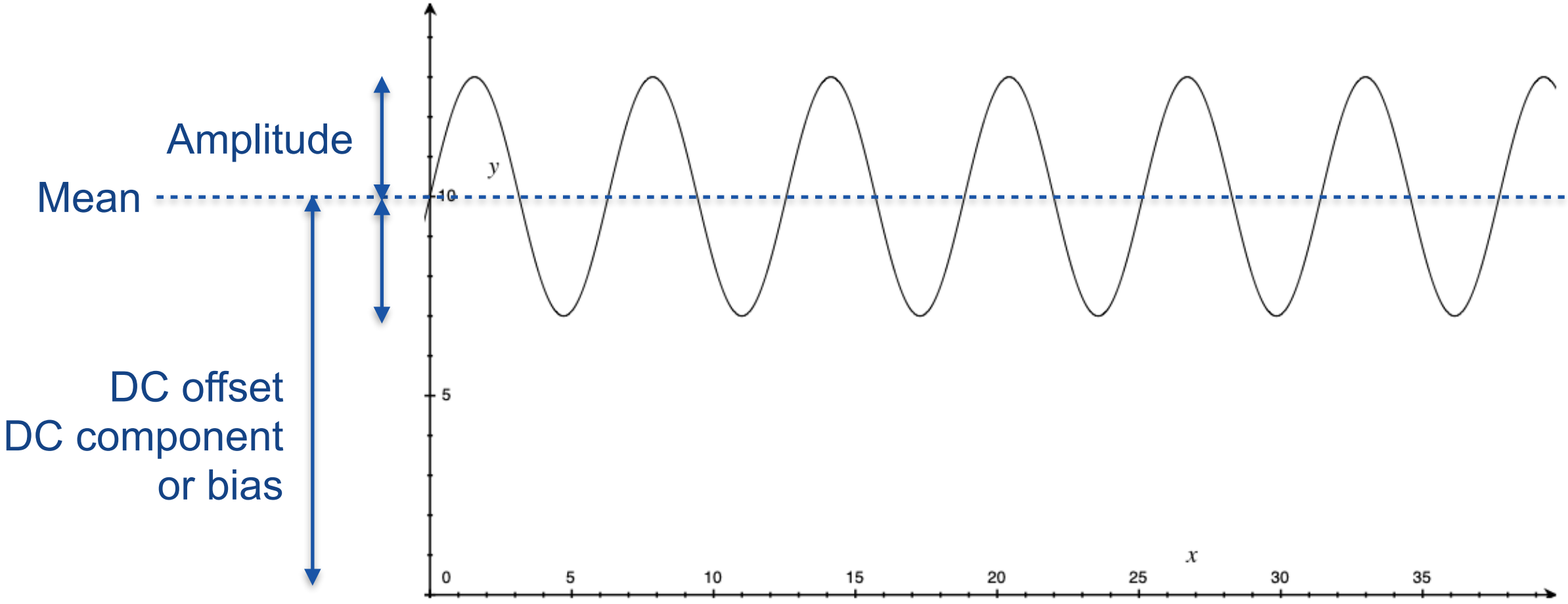


Why we need a circuit



- The signal is very weak.
- When digitalising the signal with an ADC, we need the differences to be more than a few milliVolts.
- Noise, such as power line noise (50 Hz) must be removed.
- Aliasing filter (due to problem with ADC)
- Unknown DC offset (bias)

DC offset / bias, Amplitude

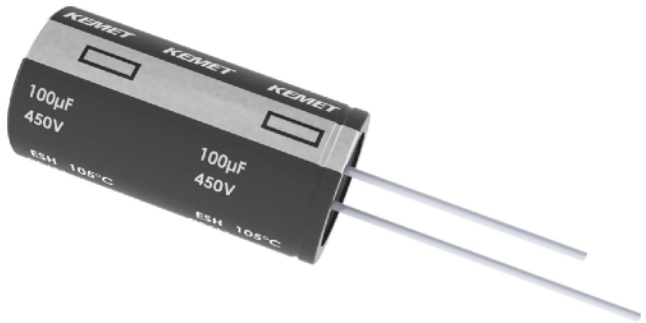


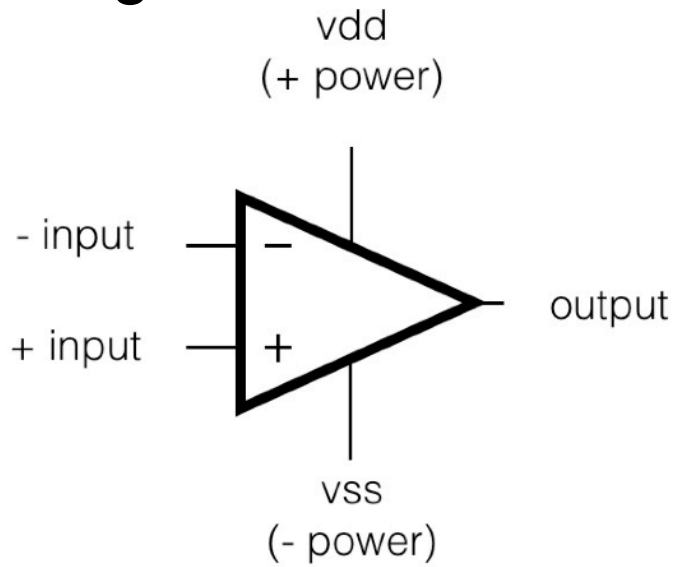
We need bias!

- A typical ADC expects an input voltage in the range

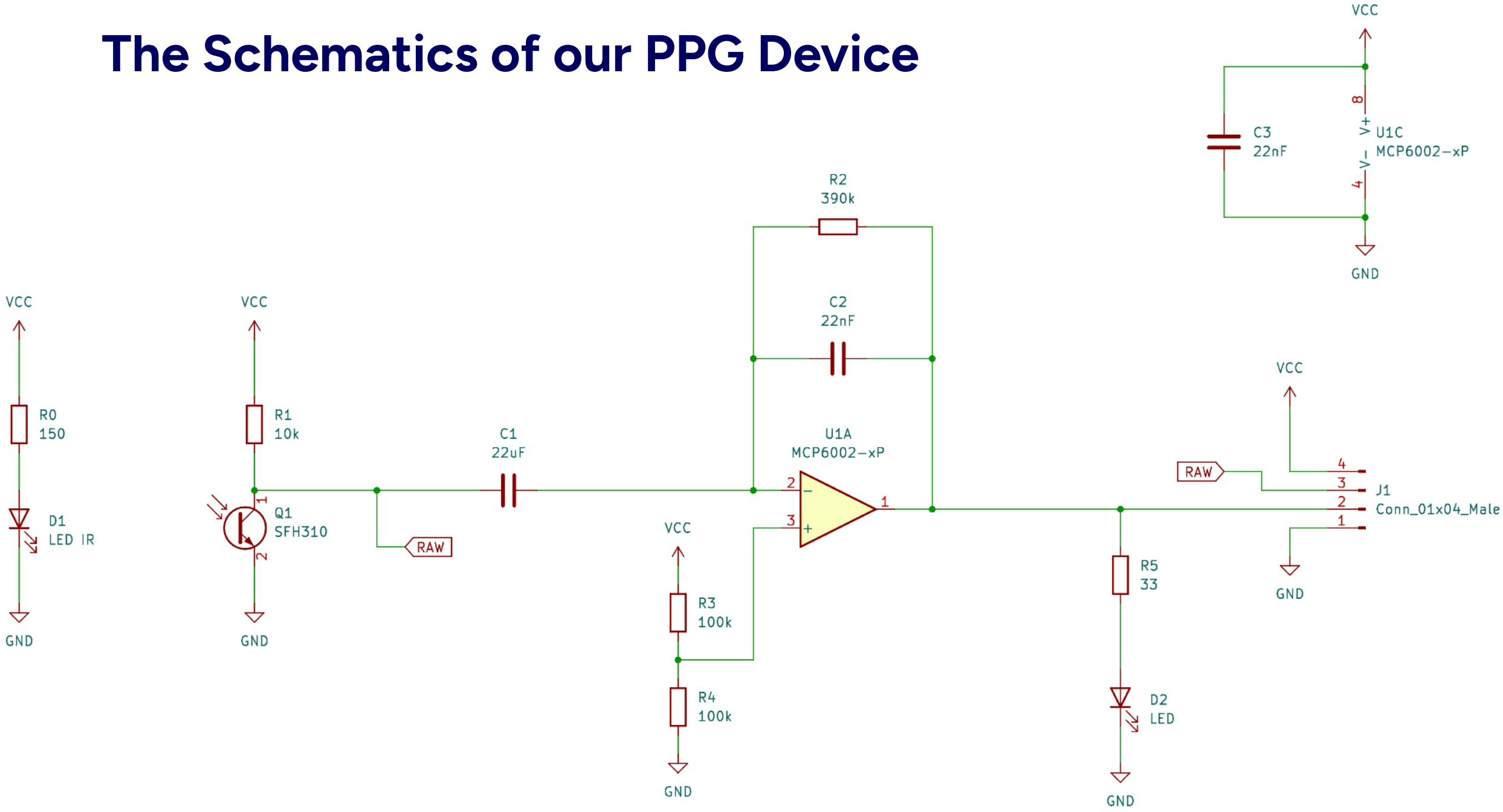
  0.0 - 3.0 Volts
- Hence, we cannot output negative voltage

  e.g. in the range of -1.5 V to +1.5 V
- Electrolyte capacitors can only work with positive voltage!
- Operational amplifiers (opamp) can only amplify to voltages within its power supply.



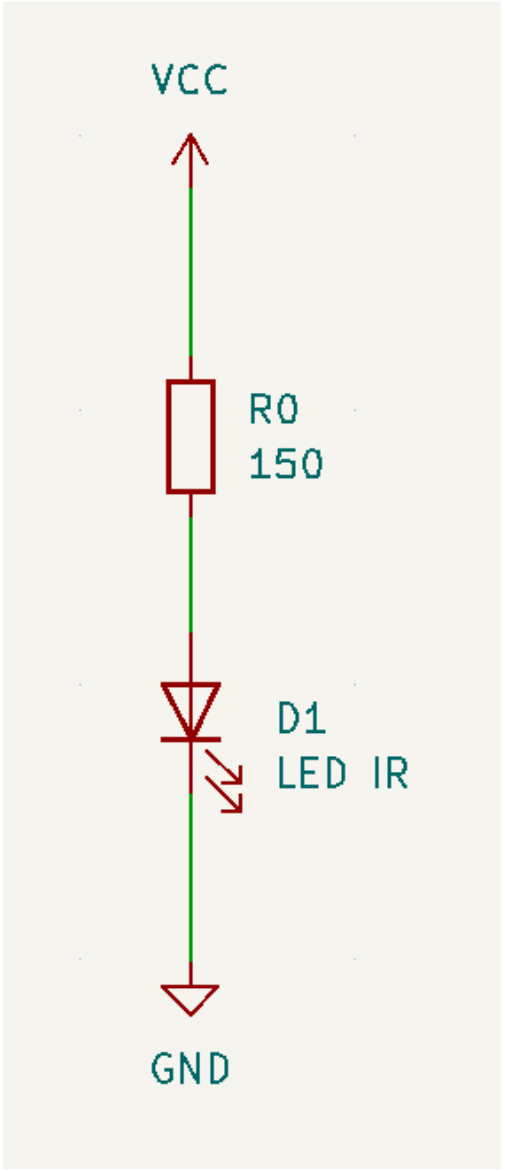


The Schematics of our PPG Device



IR LED Emitter

- 3V between VCC and GND
- Ohm's law: **U = IR**
- Gives I = 3V / 150Ω = 20 mA
- However, IR LED is not ideal. It consumes energy!
- Datasheet of Vishay TSHA5200:
- Forward voltage 1.5V.
- Hence I = 10 mA
- Below maximum forward current I_F = 100 mA



SFH310 IR transistor

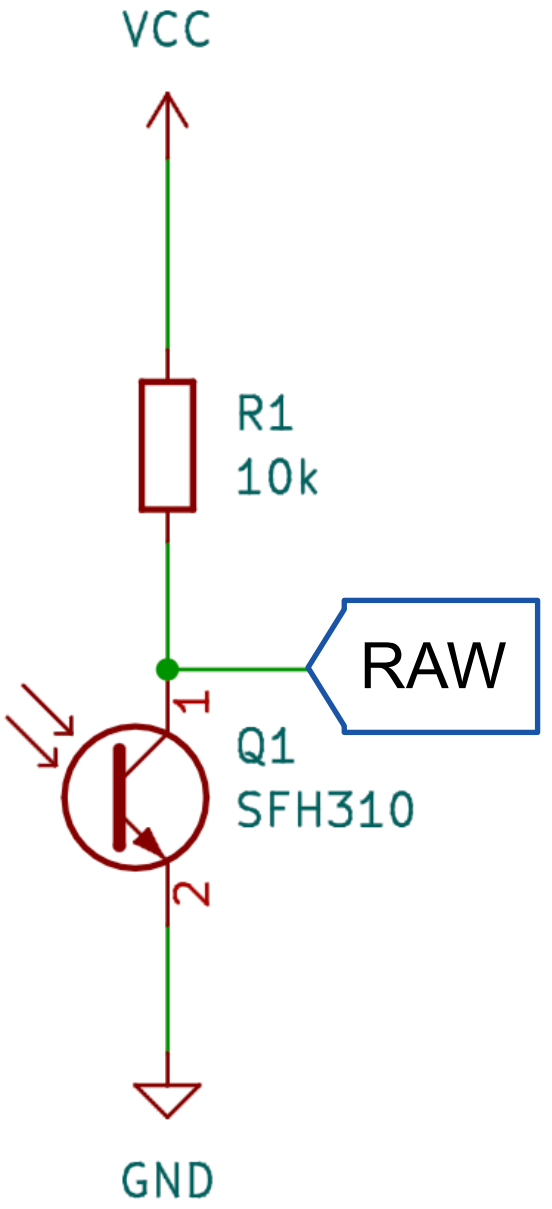
Phototransistor (Silicon NPN)

No light:

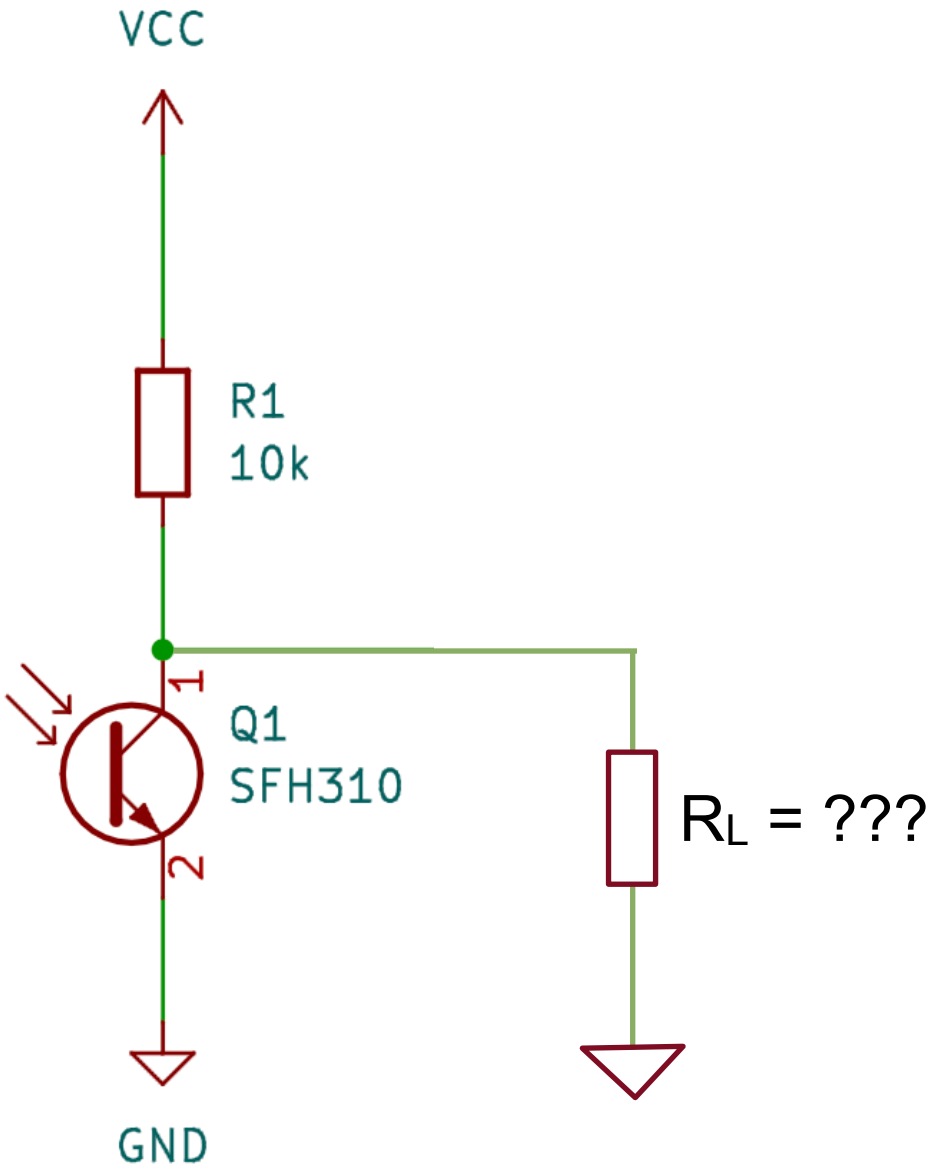
- No current (< 50 nA) through the transistor (from collector to emitter)
- Voltage over R1 is 0 V ⇒ Voltage at RAW is VCC.

Lots of light (1000 Lux):

- Current allowed through (typical 3 mA at V_{CE} = 5 V)
- R1 limits this to less than I = U/R = 0.3 mA
- The voltage between RAW and GND drops with more light



Interference with connected circuit



L = Load

If R_L is large ⇒ No problem.

Otherwise, we need to take special care!

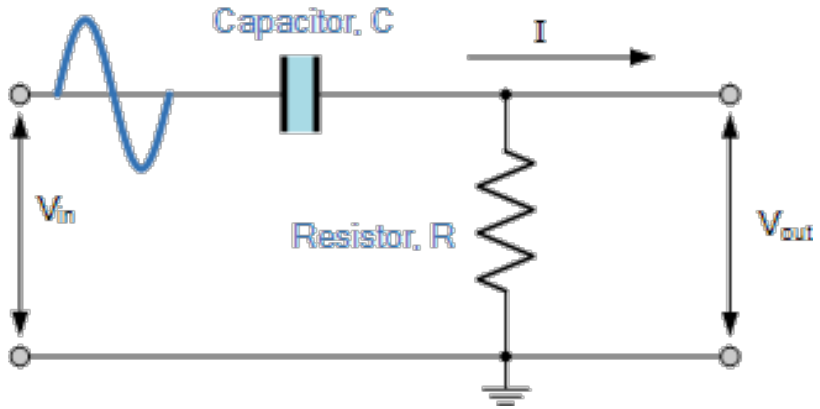
What is large?

Much larger than 10 kΩ

E.g., 1 MΩ or larger

DC offset remover (high pass filter)

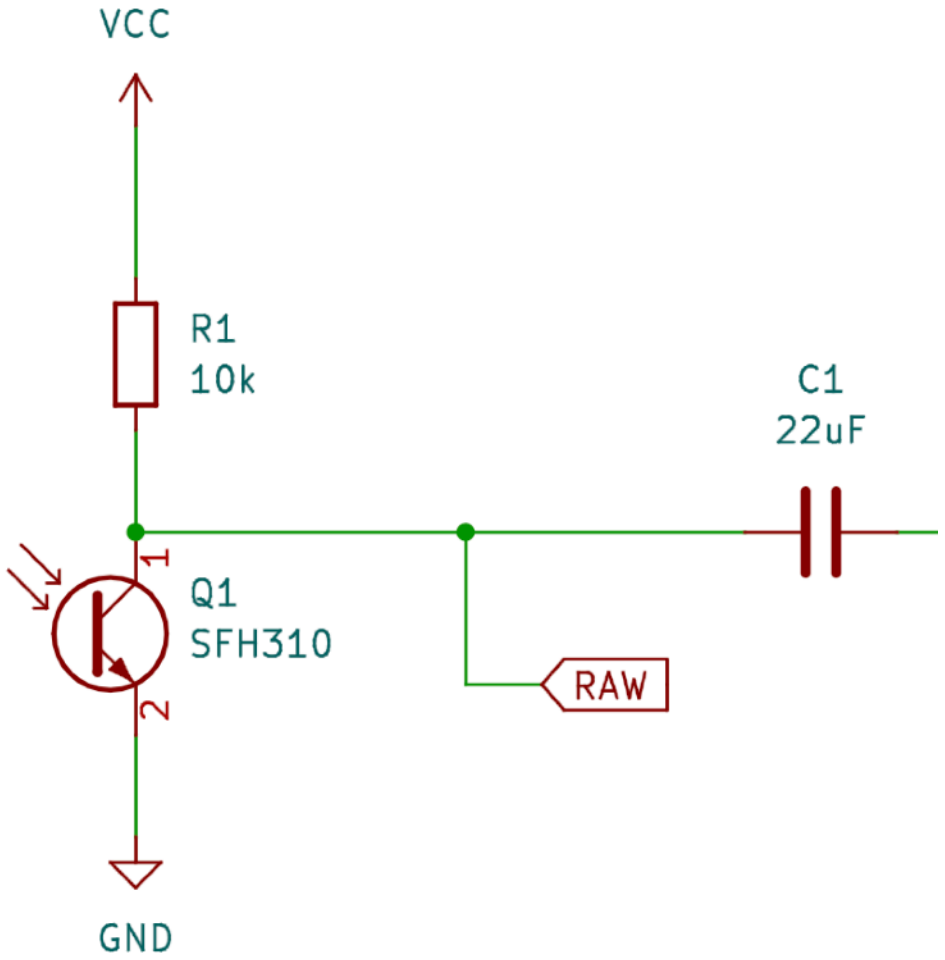
Known as a RC circuit



$$fc = \frac{1}{2\pi RC}$$

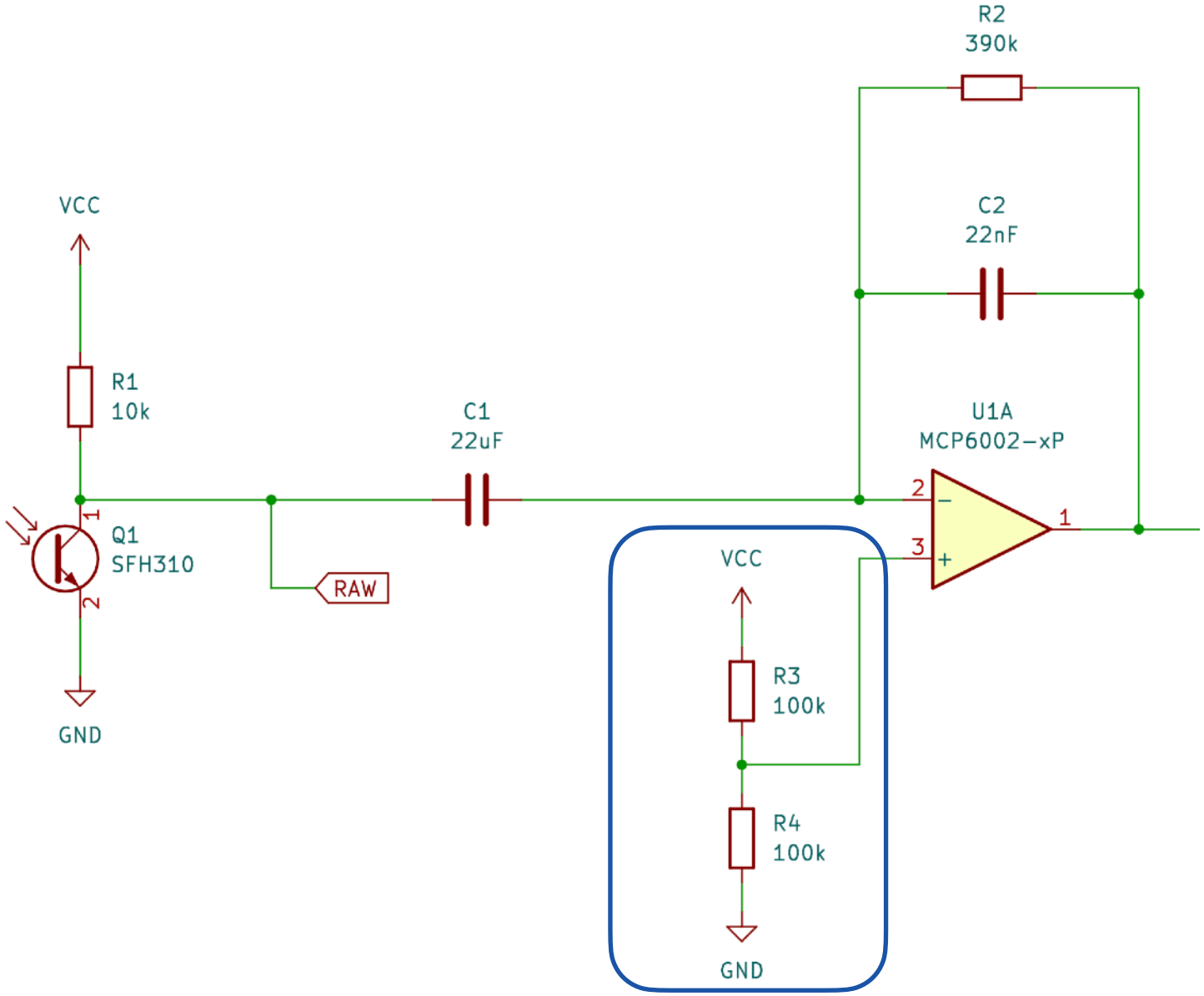
Cut-off frequency

$$Phase\ Shift\ \phi = \arctan \frac{1}{2\pi fRC}$$

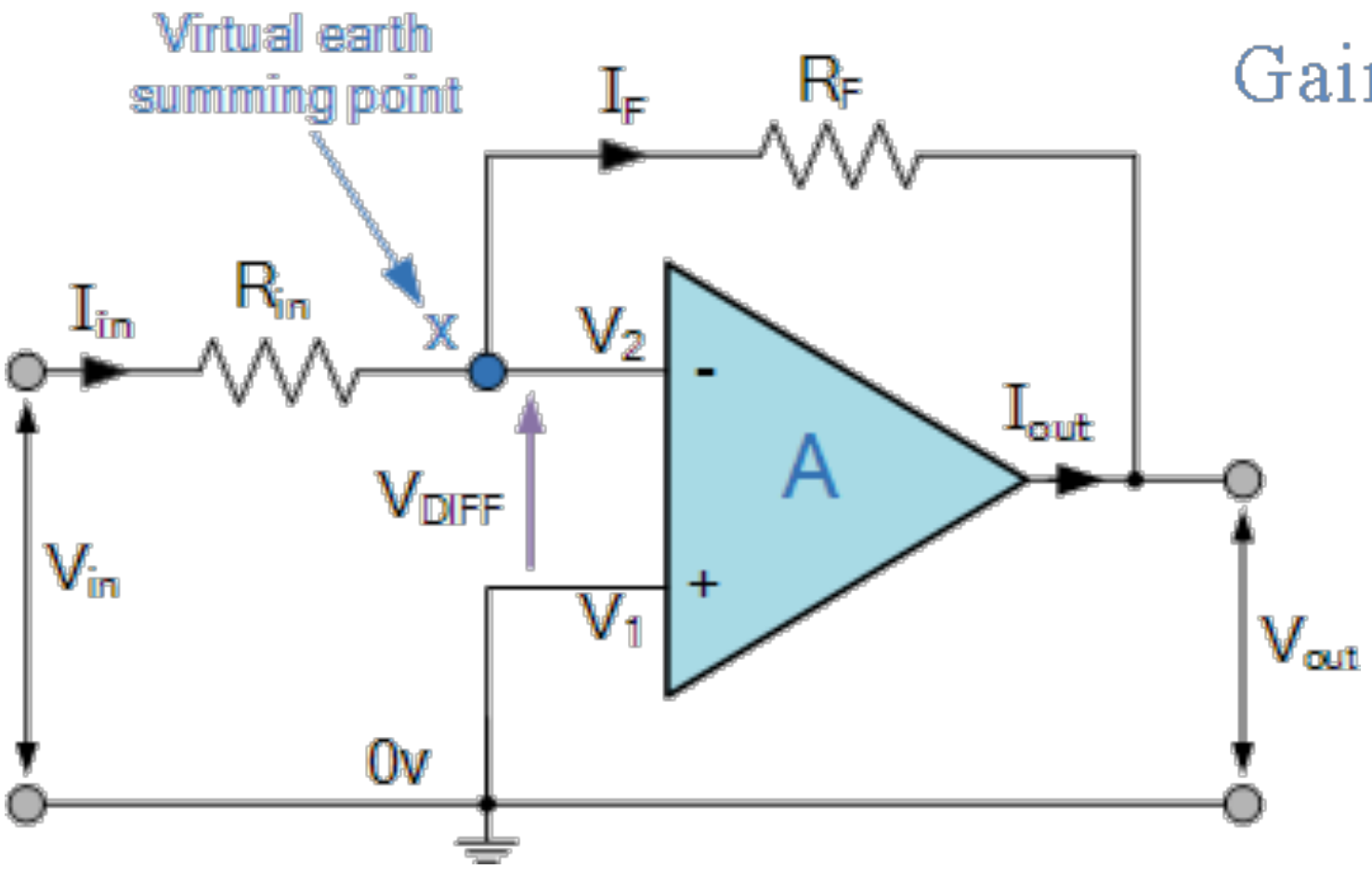


Active Band Pass Filter (BPF)

With bias



Op Amp as Amplifier (Inverting)

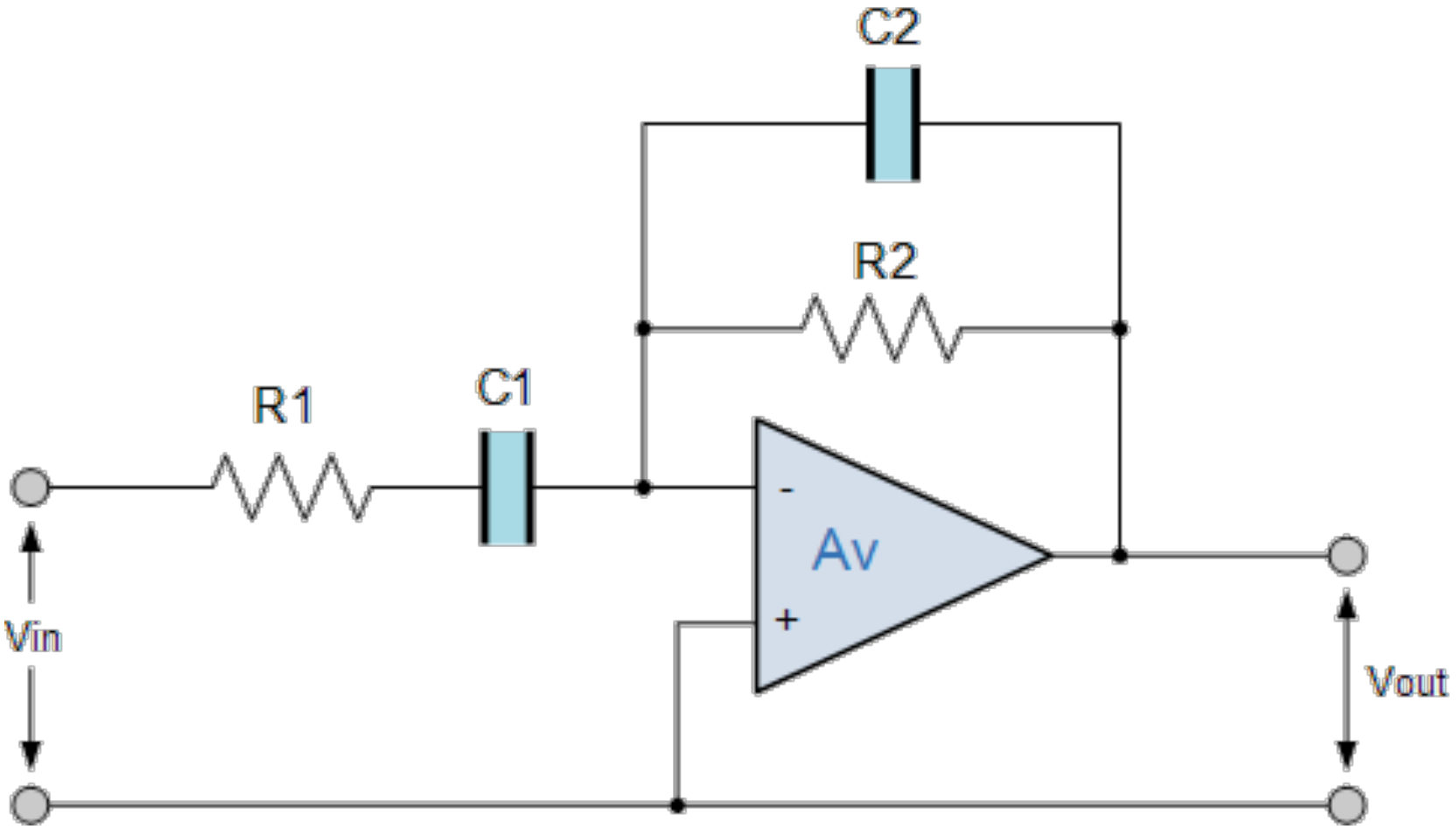


$$\text{Gain (Av)} = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_{in}}$$

V2 - V1 ≈ 0, and
No Current Flows into the Input Terminals

https://www.electronics-tutorials.ws/opamp/opamp_2.html

Op amp as band pass filter + amplifier



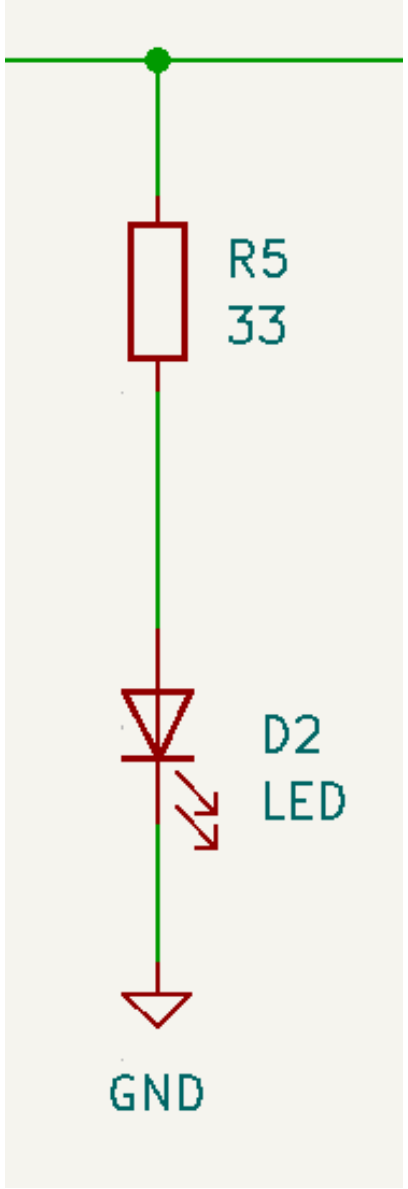
https://www.electronics-tutorials.ws/filter/filter_7.html

Blinking an LED with the pulse

- How can it blink?

- Above 1.5V, LED is on.
- Below 1.5V, LED is (almost) off.

- Typical LED forward voltage V_F = 1.7 V.
- Hence, signal less than 1.7V ⇒ almost no current through the circuit!

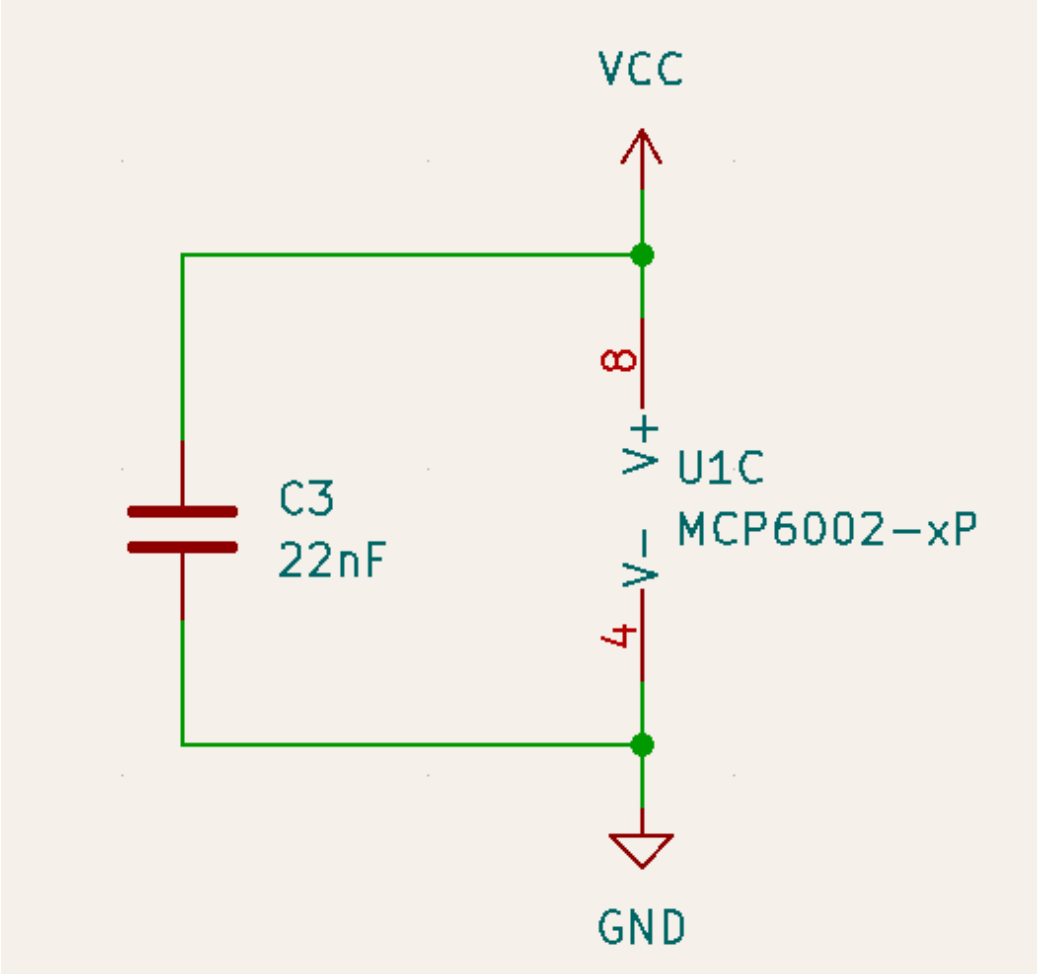


https://www.gadgetreboot.com/how-to-choose-a-resistor-for-an-led/

Op Amp Power Supply

- We supply it with 0 V and 3 V.
- Hence, no negative voltage.

- Why the capacitor?



The Tasks

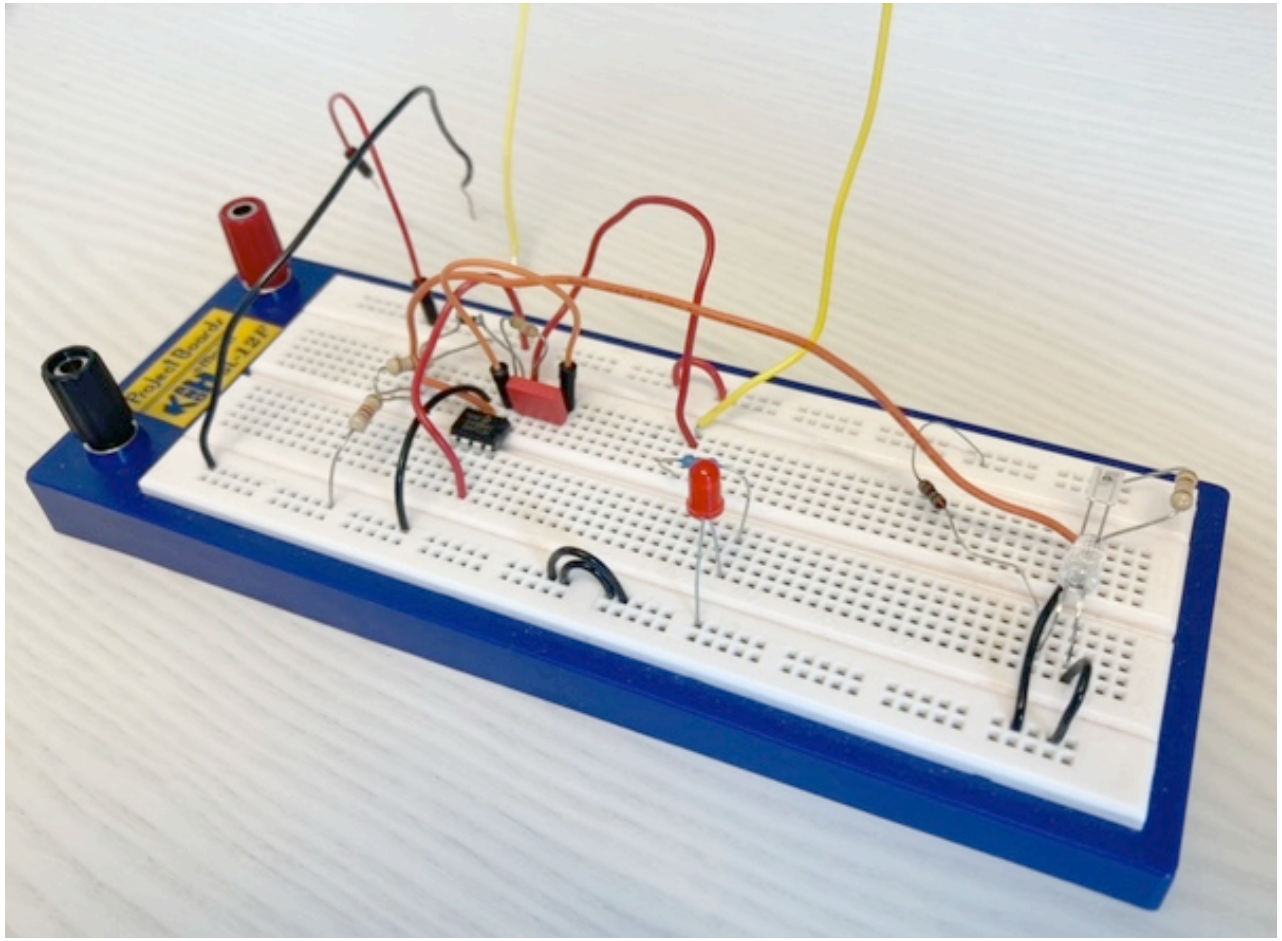
Prepare before

- By reading these links:
- https://www.gadgetreboot.com/how-to-choose-a-resistor-for-an-led/
- https://www.electronics-tutorials.ws/opamp/opamp_2.html
- https://www.electronics-tutorials.ws/filter/filter_7.html
- https://www.electrical4u.com/phototransistor/

- You may need to read other (more basic) tutorials too to understand…
- If you need to rehearse Ohm's and Kirchhoff's laws:
- https://www.youtube.com/watch?v=_rSHqvjDksg

Task 1: Connect the complete circuit on a breadboard

Top tip!
Put the wires close to the board, for better survivability in your backpack!



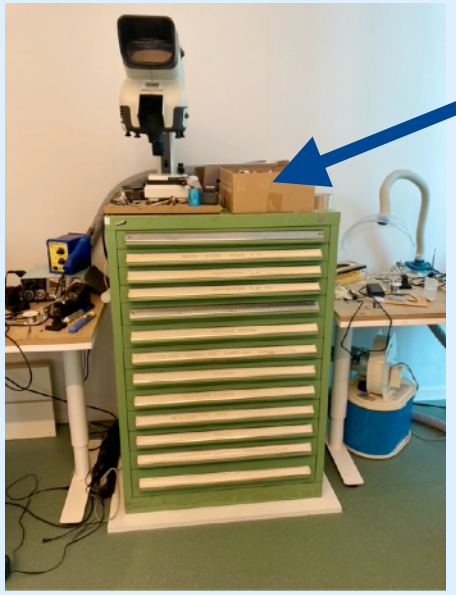
Needed for Lab 1

Find the components

- Many simple, cheap, and common components are available in MakerSpace.
- The rest are found in your kit

Or ask Linus!



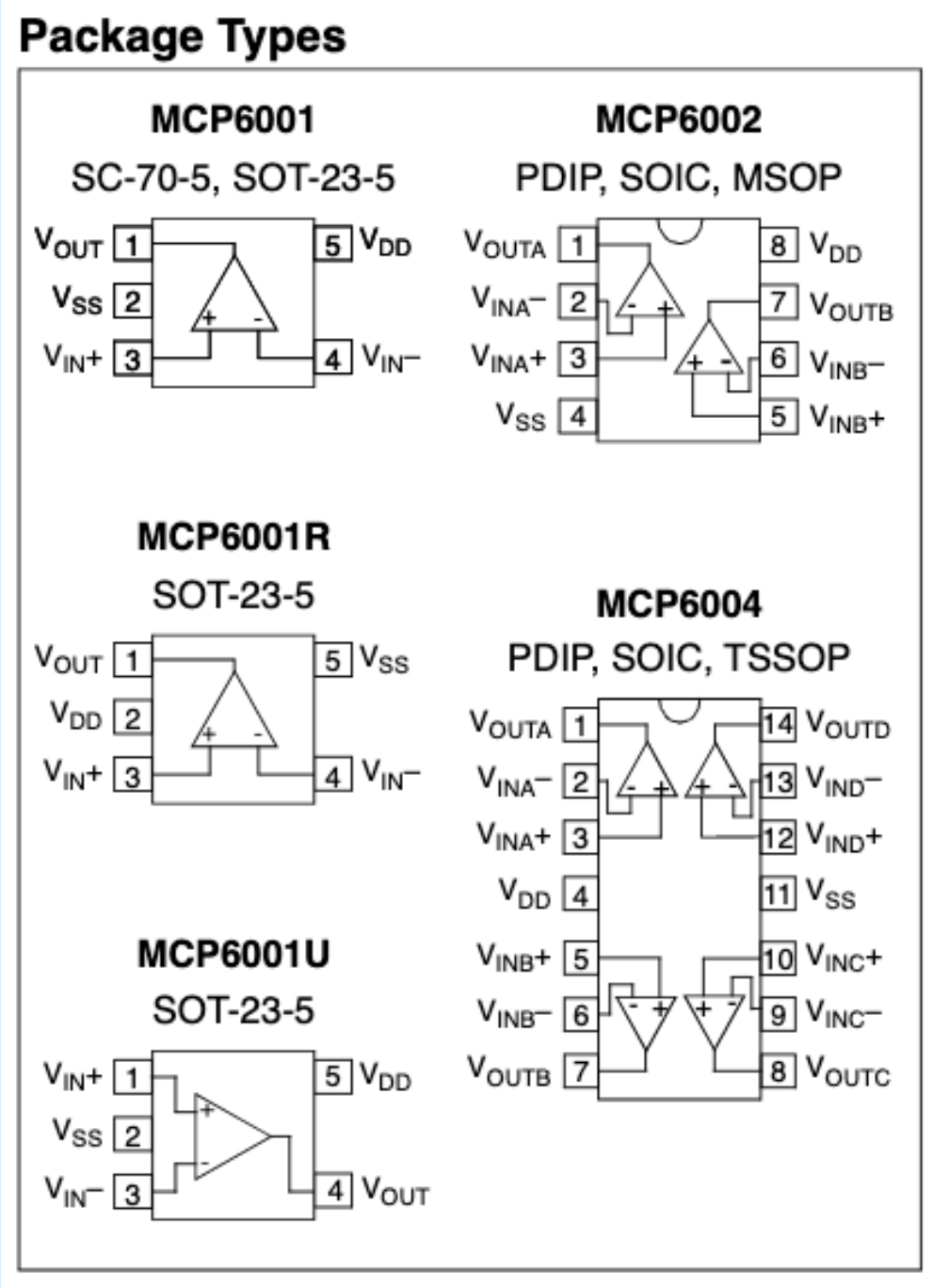




Which pin is which?

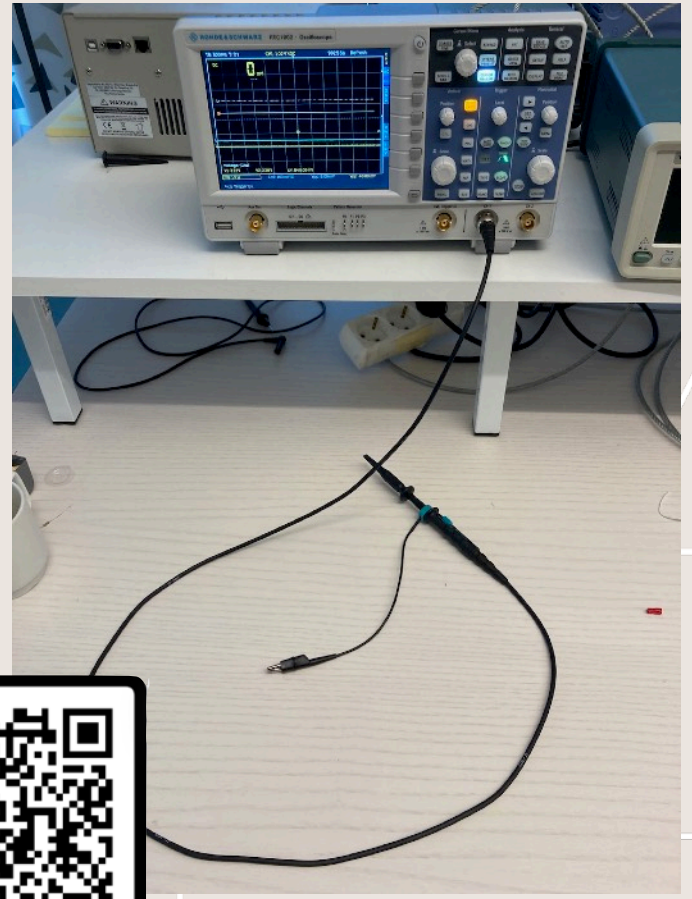
Google the *component name* and ‘datasheet’.

E.g., ‘MCP6002 datasheet’.



Test it and show to TA/teacher

1. Place a finger
2. Watch the **red** LED blink
3. Show the amplified+filtered signal on the oscilloscope

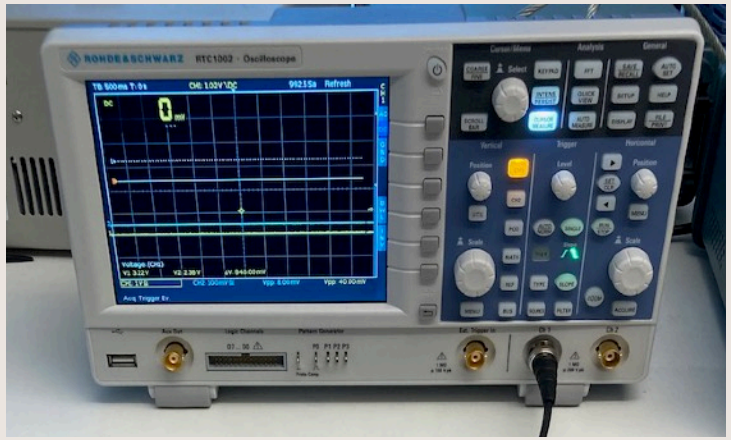


How do I use the oscilloscope?
First, use our more modern oscilloscopes by Rodhe & Schwartz RTC1002. You find the manual here:
https://www.rohde- schwarz.com/webhelp/RTC_HTML_UserManual_en/Content/welcome.htm

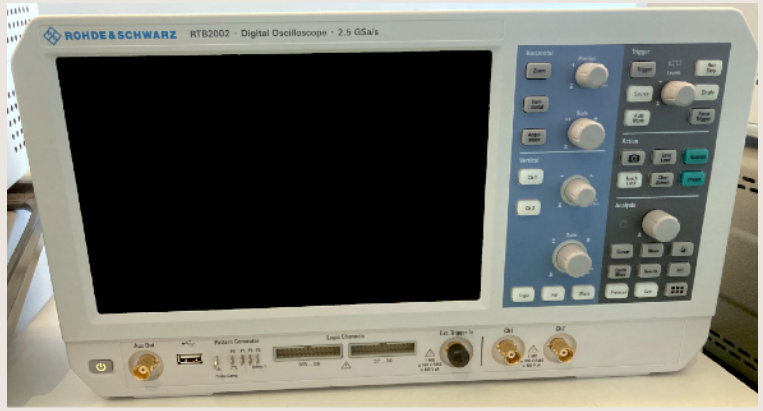
SCAN ME

Our Equipment

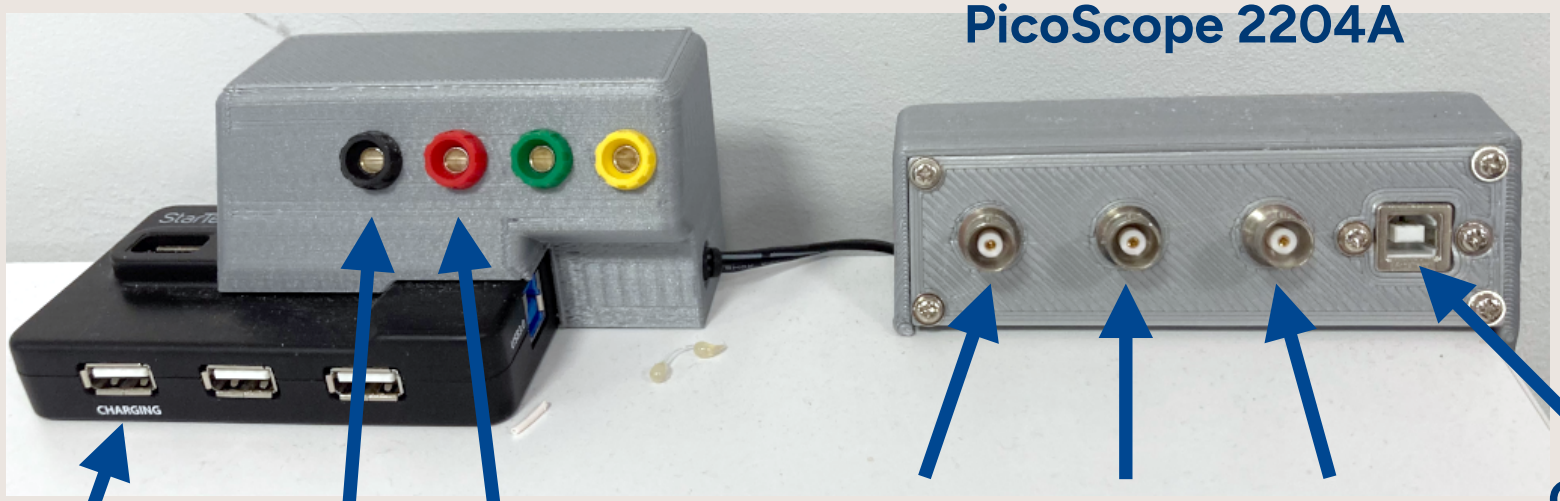
- Oscilloscope:
  - Rohde&Schwartz RTB2002
  - Rohde&Schwartz RTC1002
  - PicoScope 2204A



https://www.rohde-schwarz.com/webhelp/RTC_HTML_UserManual_en/Content/



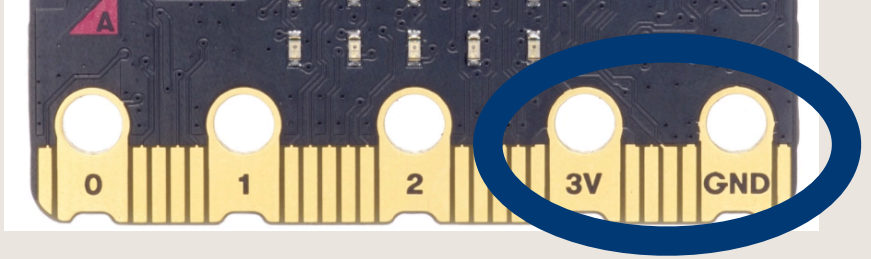
https://www.rohde-schwarz.com/webhelp/RTB2_HTML_UserManual_en/Content/welcome.htm



https://www.picotech.com/downloads

Our Equipment

Pro Tip:



- Programmable Power Supply
  - RND 320-KD3305P
- Signal Generator (arbitrary function generator)
  - Tektronix AFG1022

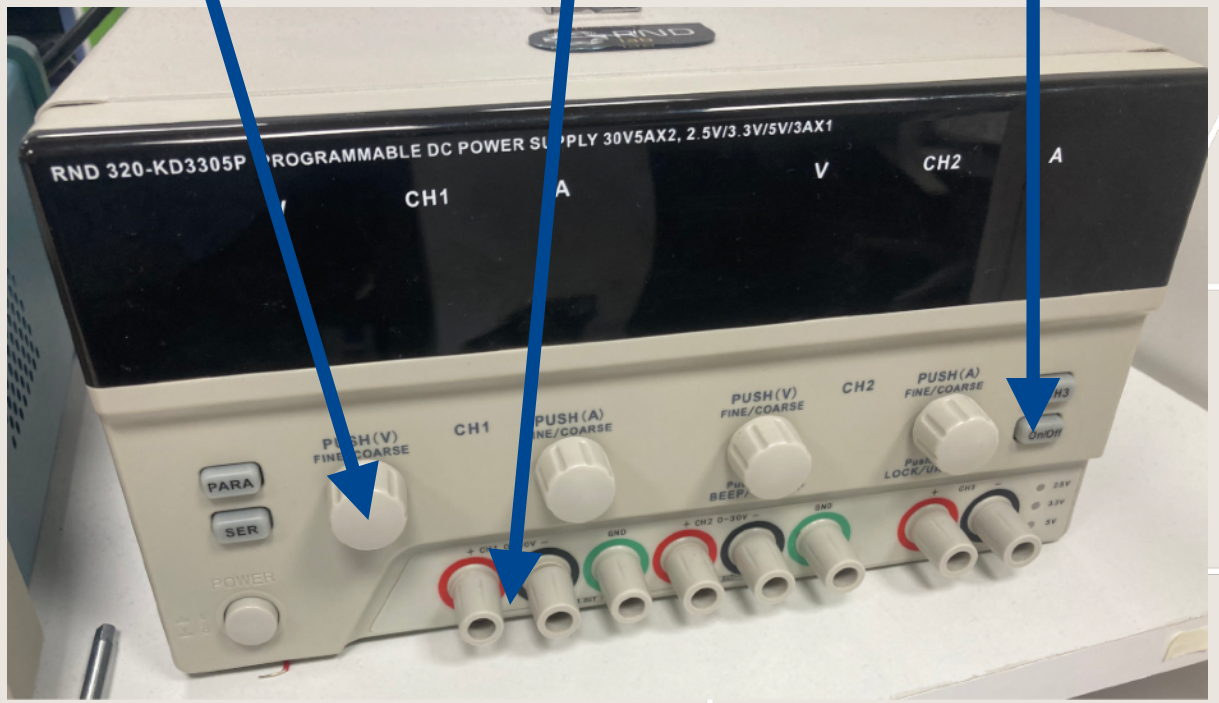
3. Press On/Off to start power supply on Ch1/2

1. Press/turn to adjust voltage

2. Connect to your circuit



https://download.tek.com/manual/AFG1022-Quick-Start-User-Manual-EN.pdf



https://www.manualslib.com/products/Rnd-320-Kd3305p-10891226.html

Calculate and simulate the band pass filter

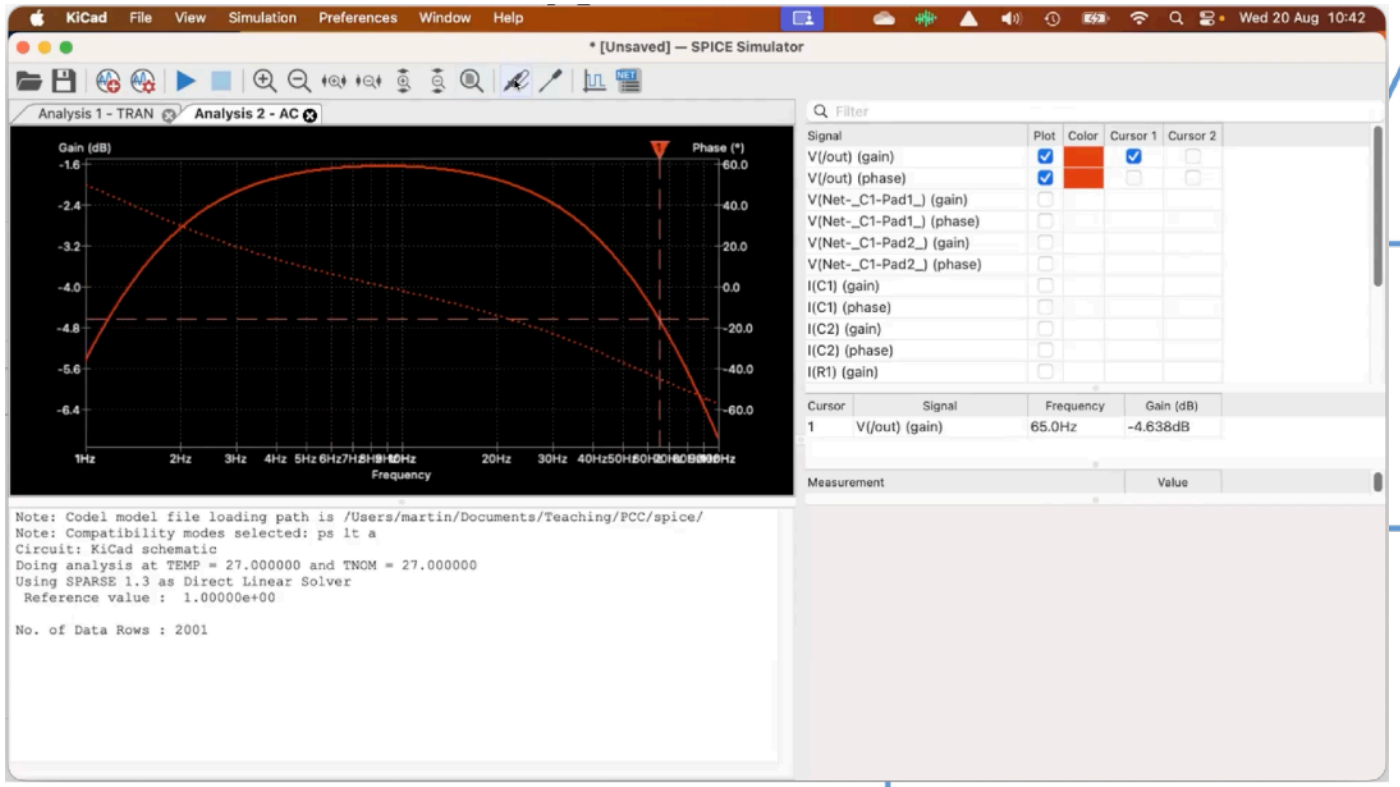
- Use the formulas here to find the lower and upper cut off frequencies:
- https://www.electronics-tutorials.ws/filter/filter_7.html

- Simulate the same thing using Multisim Live:
- https://www.multisim.com/content/P3oYMPxJVp49eu6pEfoc4b/basic-inverting-amplifier/
- Start with this circuit and extend it to the BPF used in the PPG circuit. Use the R/C values from the PPG circuit.
- **Answer the questions in the Canvas Quiz!**

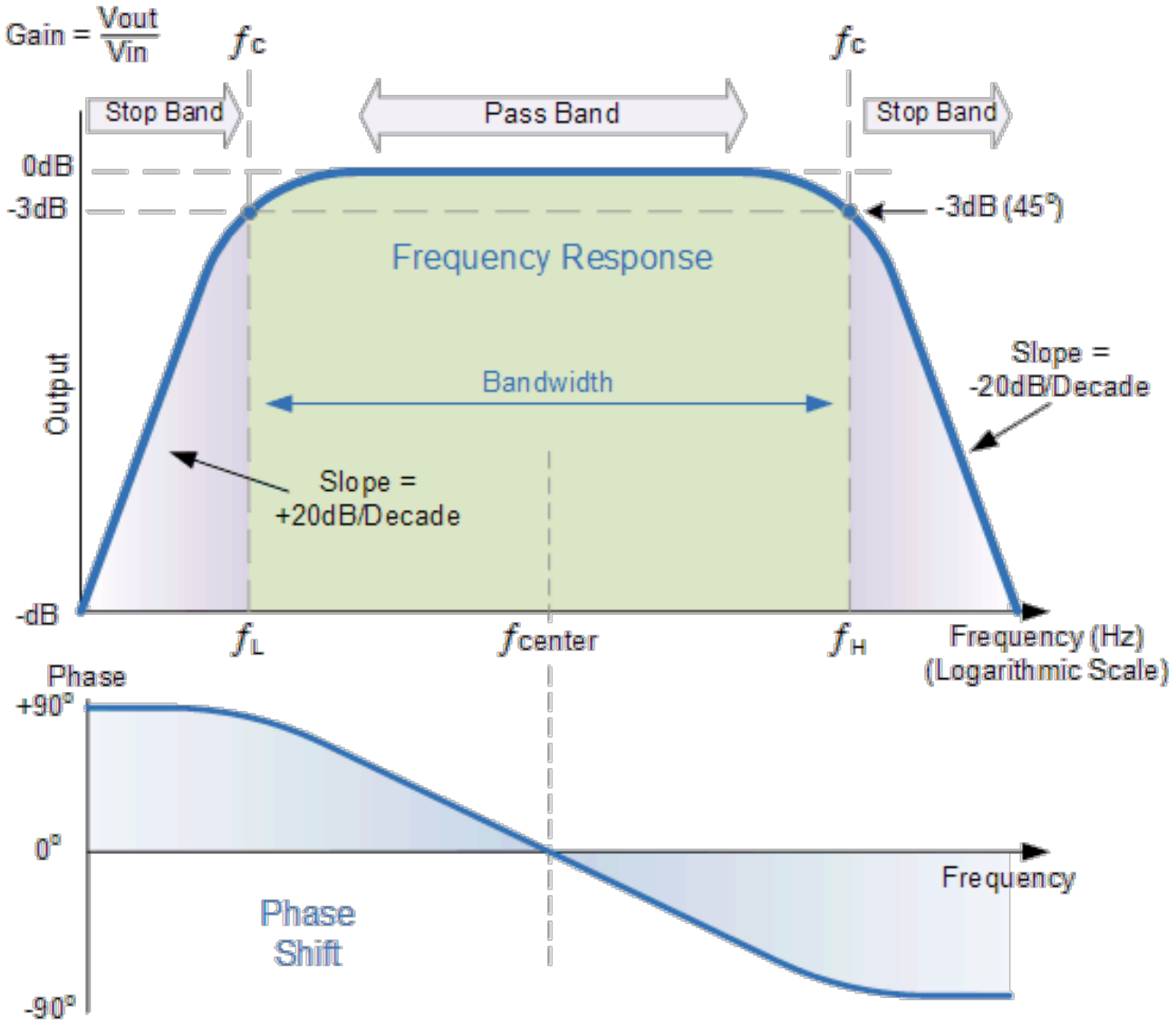
Simulate the filter + amplifier in KiCAD SPICE simulator

- Download and install KiCAD: https://www.kicad.org/download/
- Open the project files from Canvas
- Watch the tutorial: https://play.kth.se/media/SPICE+simulation+in+KiCAD+9/0_s2h8kio6

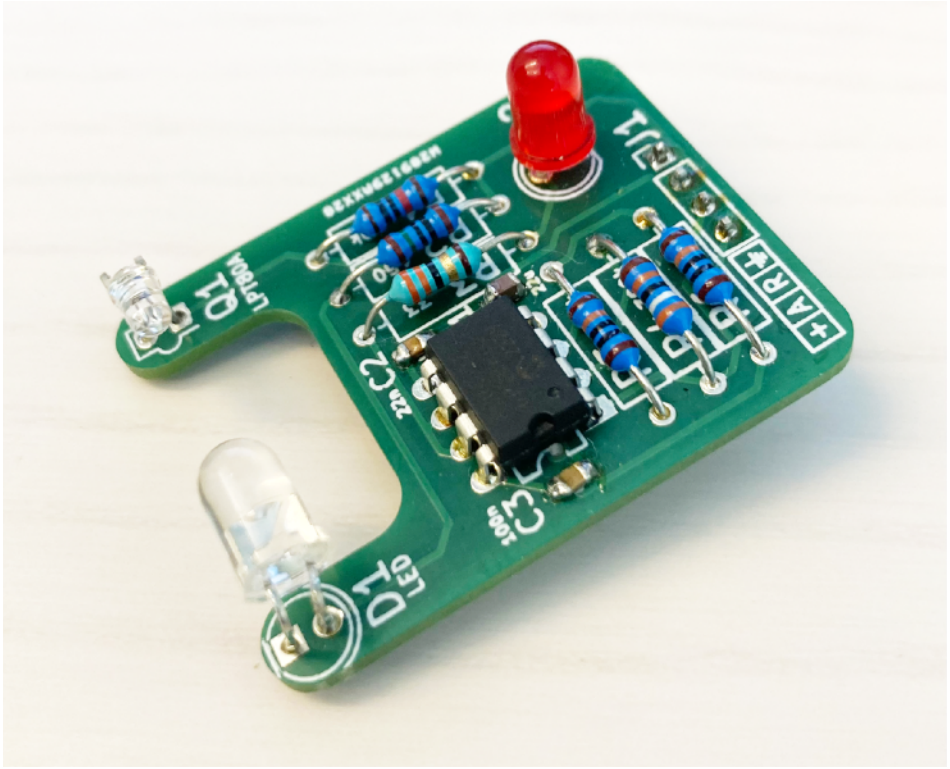
SCAN ME



Find the cut off frequencies



You may start with:



Lab 0: Build the circuit on breadboard

Lab 1: Program the device

Lab 2: PCB layout design

Lab 3: Solder

MakerSpace is too small for all of us to start with Lab 0!

Lab 1 can be started by borrowing sensors


KTH
VETENSKAP
OCH KONST